

Basic Principle

In pH measurement, the used electrode is also known as the primary battery. The primary battery is a system, whose role is to transfer chemical energy into electrical energy. The voltage of the battery is called the electromotive force (EMF). This electromotive force (EMF) is composed of two half-batteries. One half-battery is called the measuring electrode, and its potential is related to the specific ion activity; the other half-battery is the reference battery, often called the reference electrode, which is generally interlinked with the measurement solution, and connected to the measuring instrument.

pHCEM/TEM



Features

- It adopts the world-class solid dielectric and a large area of PTFE liquid for junction, non-block and easy maintenance.
- Long-distance reference diffusion channel greatly extends the service life of electrodes in the harsh environment
- It adopts PPS/PC casing and the upper and lower 3/4NPT pipe thread, so it is easy for installation and there is no need of the jacket, thus saving the installation cost.
- The electrode adopts the high-quality low-noise cable, which makes the signal output length more than 20 meters free of interference.
- There is no need for additional dielectric and there is a little amount of maintenance
- High measurement accuracy, fast responding and good repeatability.
- Reference electrode with silver ions Ag/AgCl
- Proper operation shall make service life longer.
- It can be installed in the reaction tank or pipe laterally or vertically.
- The electrode can be replaced by a similar electrode made by any other country.

Technical Parameters

Measuring range	0 - 14pH
Temperature range	0 - 60°C
Compressive strength	0.6Mpa
Slope	≥ 96%
Zero-point potential	$E_0 = 7\text{pH} \pm 0.3$
T _{response}	≤ 5s
Accuracy	better ± 0.2 pH
Internal impedance	150 - 250MΩ (25°C)
Profile	3-in-1 Electrode (Integrating the temperature compensation and the solution grounding)
Installation size	Upper and Lower 3/4NPT Pipe Thread
Connection	Low-noise cable goes out directly.
Application	Applicable to various industrial sewages, environmental protection and water treatment.